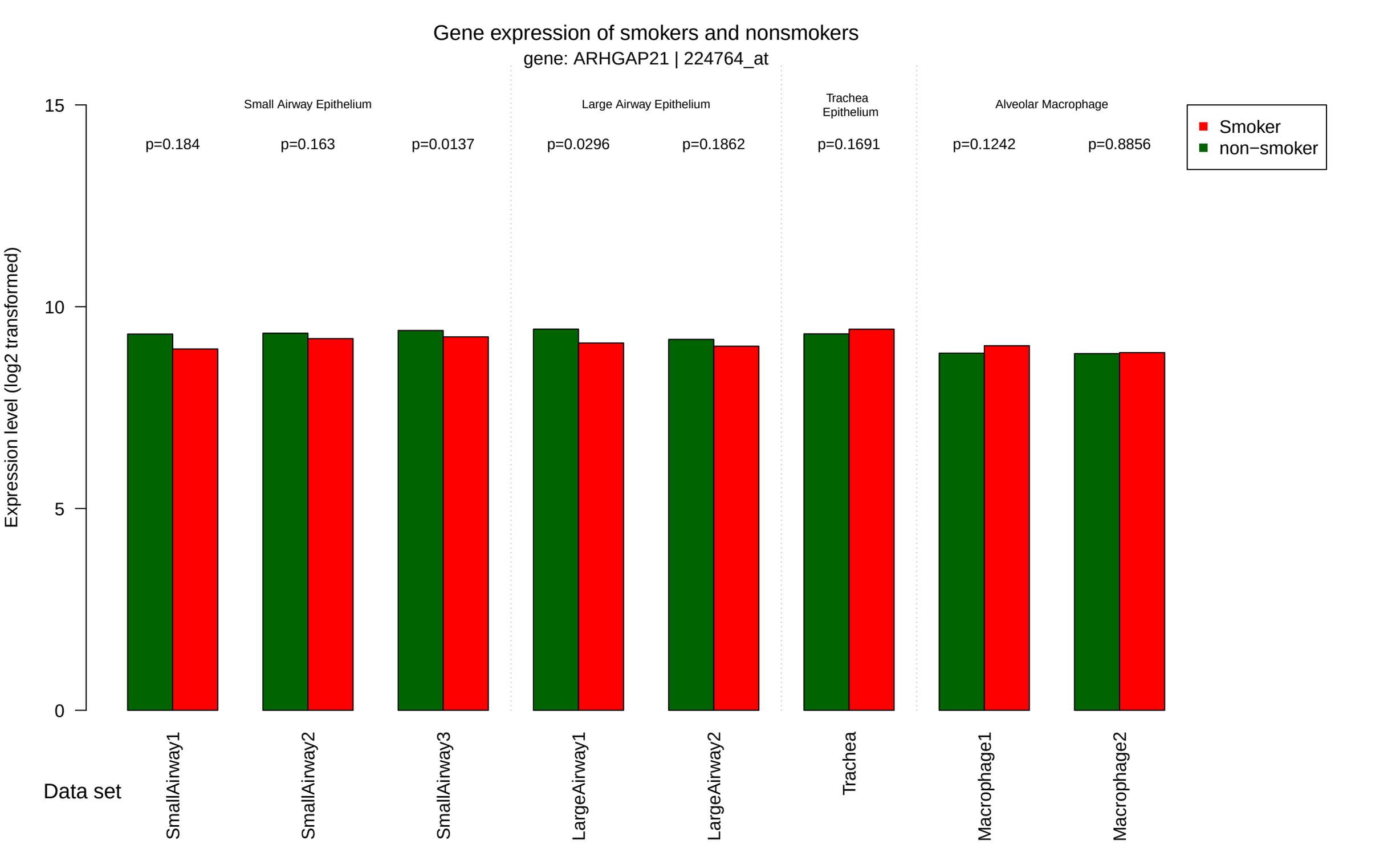
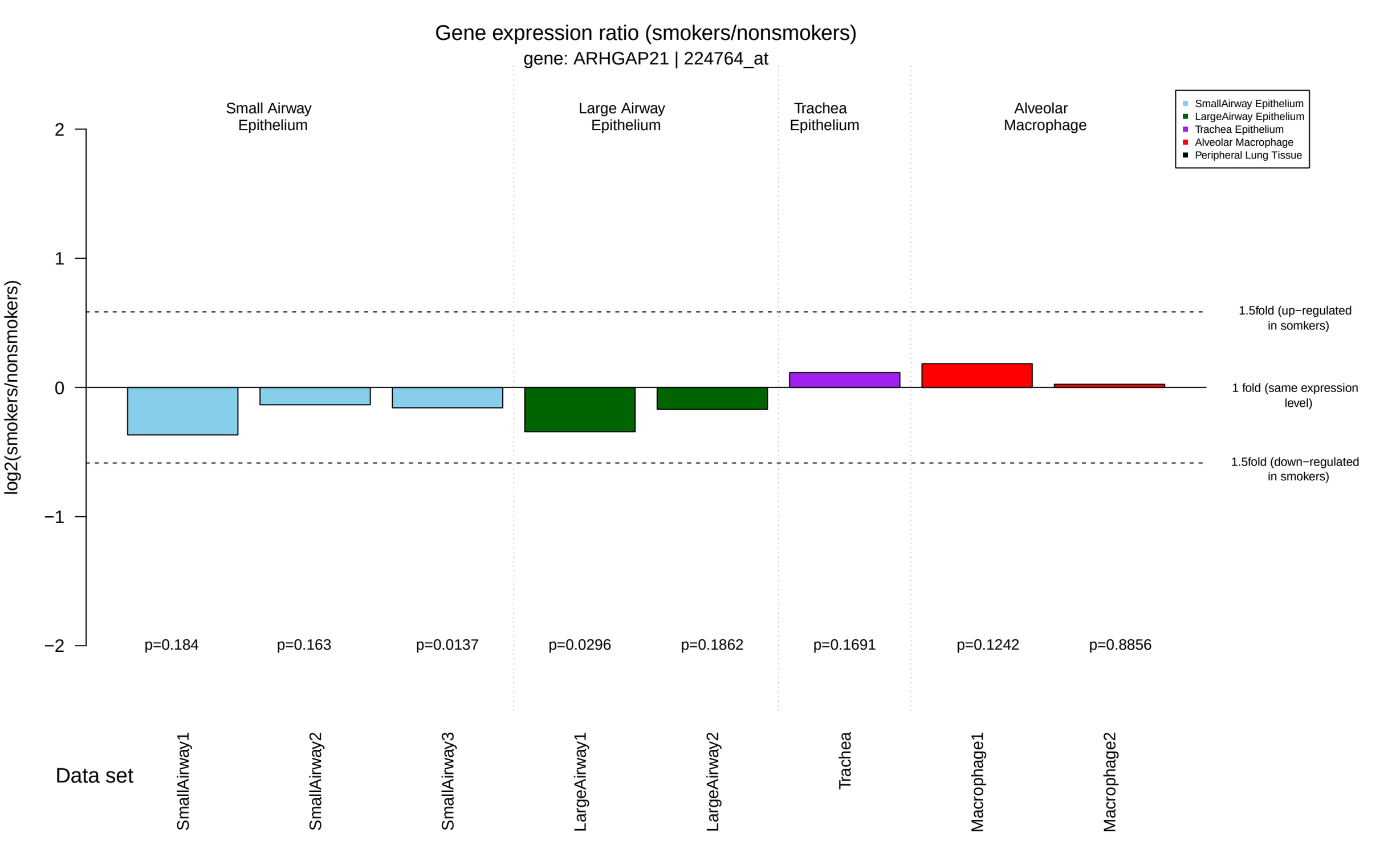
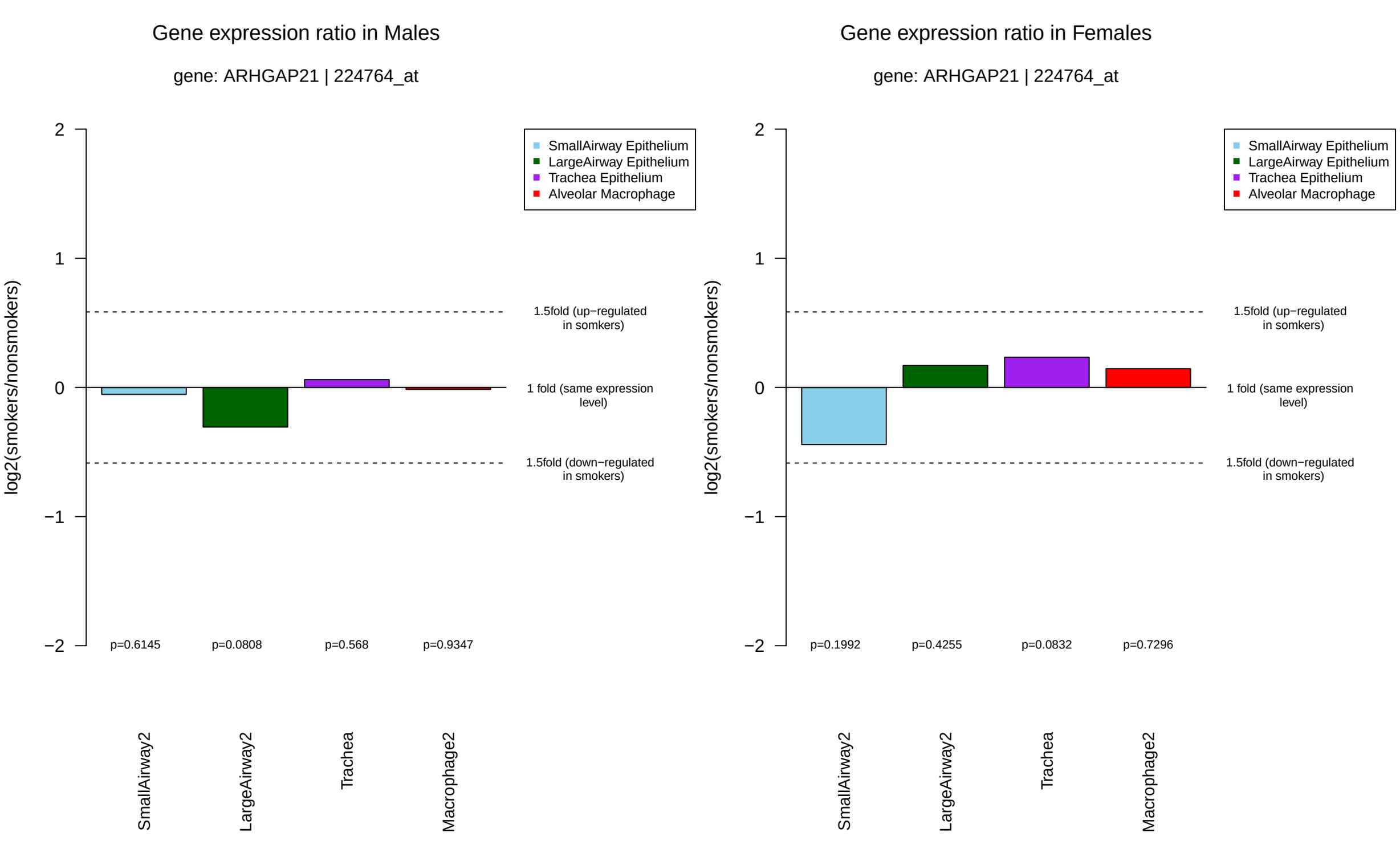
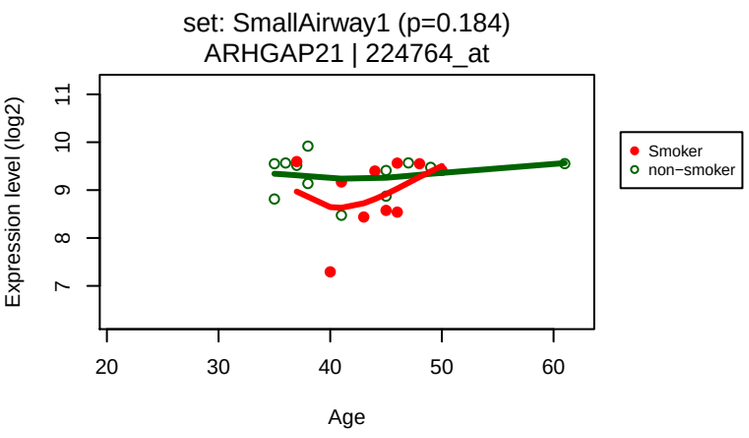
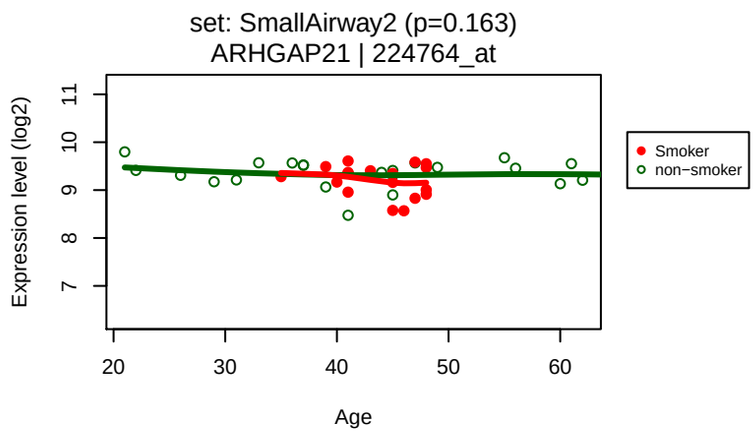
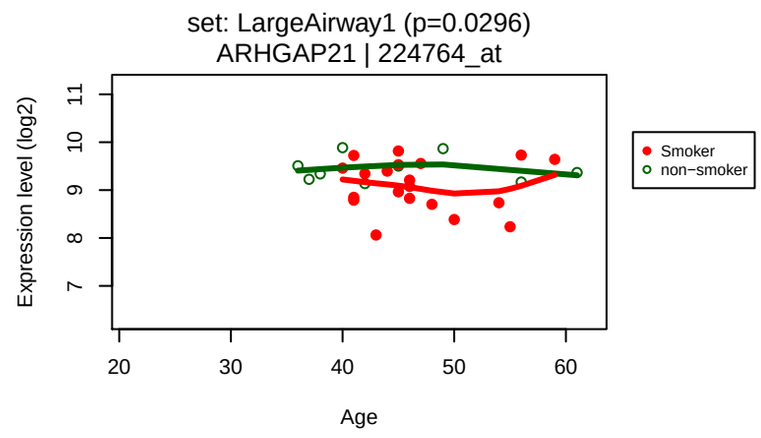

Gene expression of smokers and nonsmokers
gene: ARHGAP21 | 224764_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.184
p=0.163
p=0.0137
p=0.0296
p=0.1862
p=0.1691
p=0.1242
p=0.8856
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

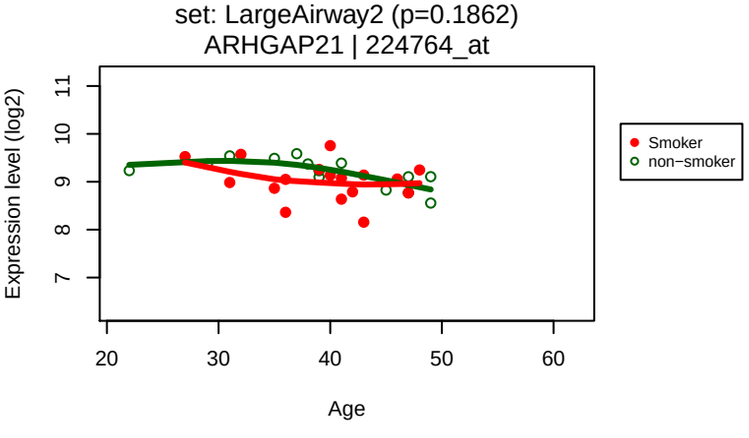

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP21 | 224764_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.184
p=0.163
p=0.0137
p=0.0296
p=0.1862
p=0.1691
p=0.1242
p=0.8856
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

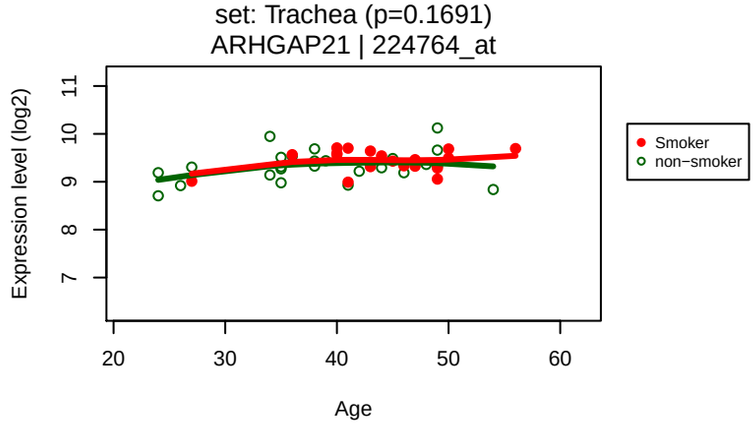

Gene expression ratio in Males
gene: ARHGAP21 | 224764_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6145
p=0.0808
p=0.568
p=0.9347
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP21 | 224764_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1992
p=0.4255
p=0.0832
p=0.7296
SmallAirway2
LargeAirway2
Trachea
Macrophage2

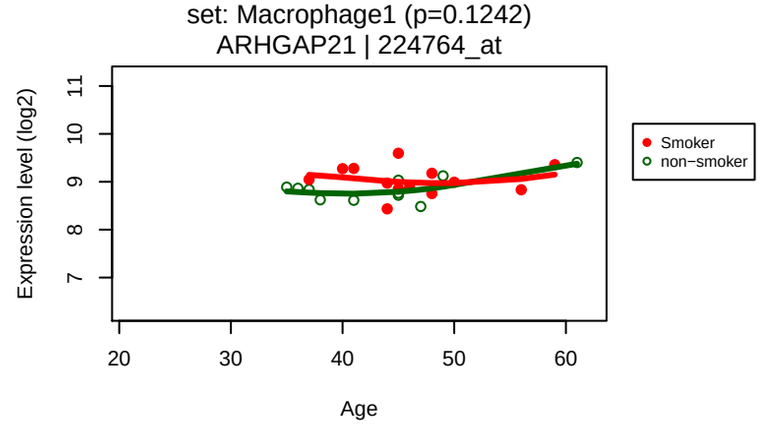

set: SmallAirway1 (p=0.184)
ARHGAP21 | 224764_at
Expression level (log2)
Age
Smoker
non−smoker

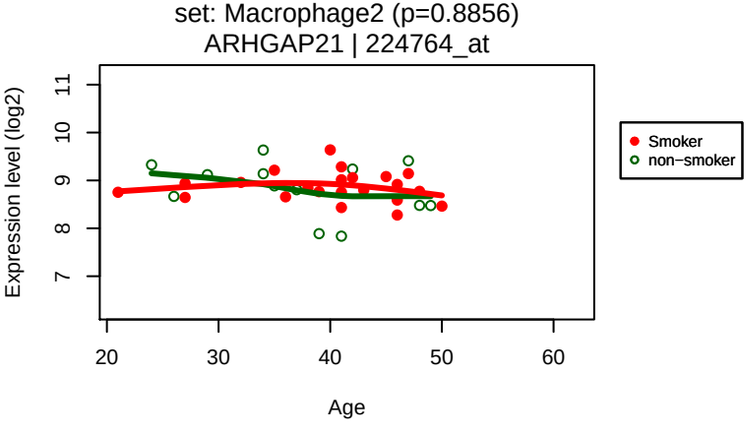

set: SmallAirway2 (p=0.163)
ARHGAP21 | 224764_at
Expression level (log2)
Age
Smoker
non−smoker

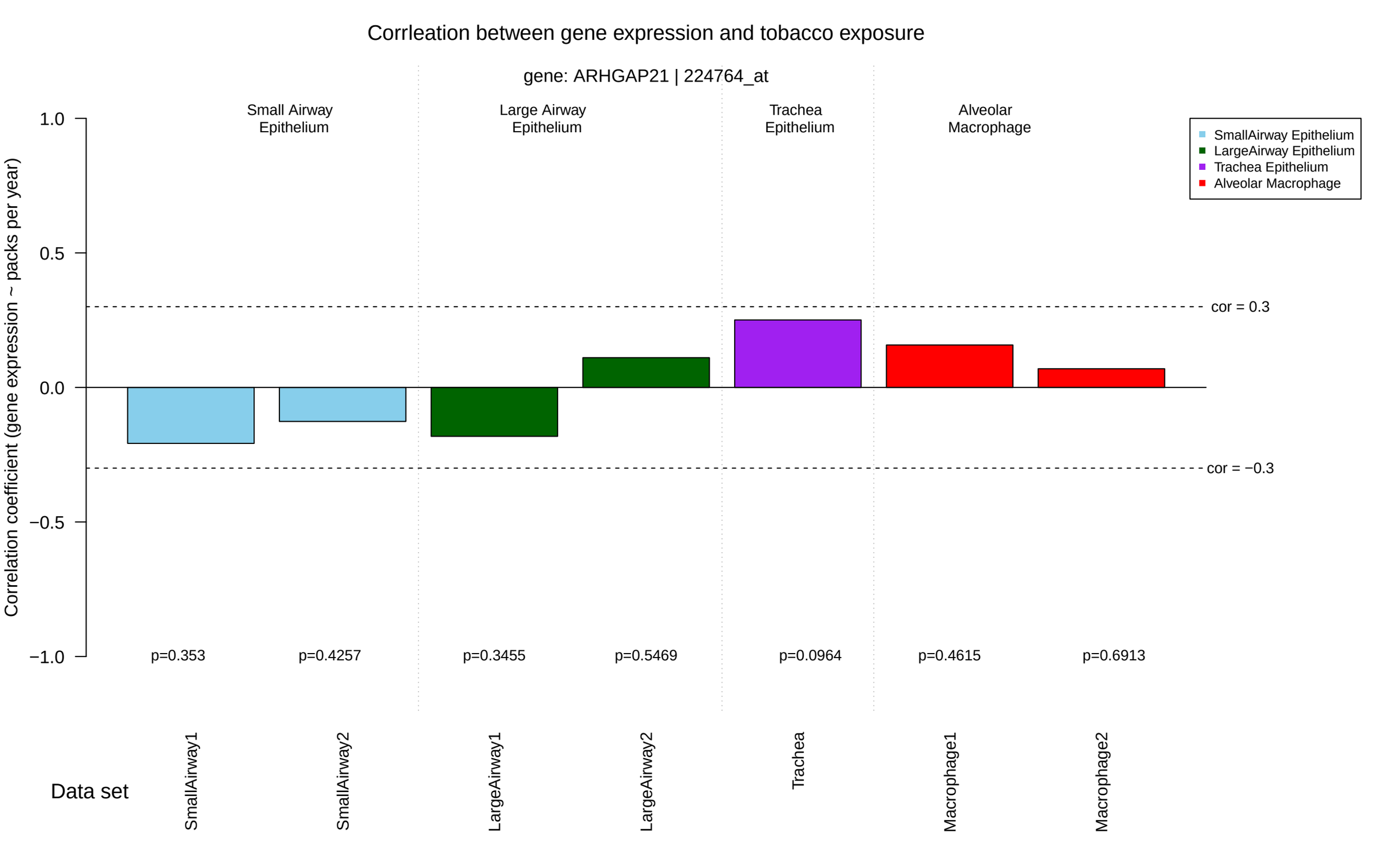

set: LargeAirway1 (p=0.0296)
ARHGAP21 | 224764_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1862)
ARHGAP21 | 224764_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1691)
ARHGAP21 | 224764_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1242)
ARHGAP21 | 224764_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8856)
ARHGAP21 | 224764_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP21 | 224764_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.353
p=0.4257
p=0.3455
p=0.5469
p=0.0964
p=0.4615
p=0.6913
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2